



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 2: PROJECT MANAGEMENT SYSTEM ANALYSIS & DESIGN (SECP2613-02)

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Question 1

a. How would you assess the feasibility of the systems project Ahmad is proposing?

Based on conversation between Abu and Ahmad, the systems proposed by Ahmad can be feasible as all the customers are satisfied with the newer systems after the interview as the newest system involves the image update that can make every user excited.

b. Based on what Abu has said about the managers, users, and systems people, what seems to be the operational feasibility of the proposed project?

In terms of operational feasibility, it is whether the users want to use the system or not. Based on what Ahmad interview notes, it seems that the manager, the users and the people who have extensive contact with the system have been approved to pursue the project. Therefore, the newer system can be implemented.

c. What about the economic feasibility?

In terms of economic feasibility, a conversation between Ahmad and Abu talking about the cost of making a new system is possibly high and an uncertain amount in investment. For example, to develop a new or to upgrade the system, they should consider development costs, training costs, maintenance costs, risk management costs and opportunity costs.

d. What about the technological feasibility?

Regarding the technological feasibility, although some users are satisfied with the current system, it is better to make an upgrade to the latest technology as the technology is developing from time to time. Furthermore, the operation or project with the newest system will be efficient and smoothly.

e. Based on what Ahmad and Abu have discussed, would you recommend that a full-blown systems study be done? Discuss your answer in a paragraph.

Based on discussion between Ahmad and Abu, I recommend that a full-blown systems study needs to be done because even if the cost or time is uncertain to develop a new system, the latest technology is much better and easier to all users including a manager, system analyst, employee and customer. Learning or experiencing something new is good to keep life much easier and ready to face any challenge in the future.

Question 2

Think of all the tasks that you perform when you want to propose a new computer system for student registration. Include any research, decisions, or financial issues that are related to the development of a new computer system. Create a Work Base Schedule (WBS) that shows all the tasks, their estimated duration, and any predecessor tasks.

1.	Planning	
1.1.	Determine scope of the project	
1.1.1.	Predecessors: None	
1.1.2.	Duration: 1 days	
1.2.	Determine the job scope for each person	
1.2.1.	Predecessors: None	
1.2.2.	Duration: 2 days	
2.	Analysis	
2.1.	Identify the needs assessment	
2.1.1.	Predecessors: 1.1 & 1.2	
2.1.2.	Duration: 2 days	
2.2.	Obtain the requirement	
2.2.1.	Predecessors: 2.1	
2.2.2.	Duration: 3 days	
3.	Design	
3.1.	Create design	
3.1.1.	Predecessors: 2.2	
3.1.2.	Duration: 1 weeks	
3.2.	Defining overall system architecture	
3.2.1.	Predecessors: 3.1	
3.2.2.	Duration: 2 weeks	
4.	Development	
4.1.	Start the development	
4.1.1.	Predecessors: 3.2	
4.1.2.	Duration: 3 weeks	
4.2.	Coding phase	
4.2.1.	Predecessors: 3.2	
4.2.2.	Duration: 3 weeks	
5.	Testing	
5.1.	Debugging phase	
5.1.1.	Predecessors: 4.1 & 4.2	
5.1.2.	Duration: 2 weeks	
5.2.	Fixed bug	
5.2.1.	Predecessors: 5.1	
5.2.2.	Duration: 2 weeks	
5.3.	Testing system	
5.3.1.	Predecessors: 5.2	
5.3.2.	Duration: 3 days	
6.	Deployment	
6.1.	Planning the deployment of the system	

6.1.1.	Predecessors: 5.3
6.1.2.	Duration: 5 days
6.2.	Deploy the system
6.2.1.	Predecessors: 6.1
6.2.2.	Duration: 1 weeks
7.	Maintenance
7.1.	Provide update
7.1.1.	Predecessors: 6.2
7.1.2.	Duration: Ongoing
7.2.	User feedback
7.2.1.	Predecessors: 6.2
7.2.2.	Duration: Ongoing

Question 3

Many of today's projects involve team members scattered across different time zones and in different physical locations. Moreover, the projects may have adopted an agile methodology, which reduces cycle time dramatically. Write a brief report that summarizes some of the key differences a manager would face managing this type of project, as opposed to a traditional project.

Introduction:

A project needs to be handled correctly. It is to prevent the project from failing. Therefore, someone needs to manage the project from the beginning until it succeeds. So, the manager plays such a big role in handling a project. They can follow System Development Life Cycle (SDLC) to make it easier. But, they need to choose a methodology which makes their work better. It is either Traditional methodology or the famous today which is Agile methodology. The report is going to talk about key differences between this both methodology.

Key difference

→ Communication

For traditional methodology, because it is formal, the meetings are often planned. So, it is difficult to break or change the protocol that was already planned. This will be distributed with stakeholders. On the other hand, for agile methodology, it is informal. Usually there will be daily stand-up meetings. This is going to make interaction easier among team members and stakeholders.

→ Flexibility

It is less flexible for traditional methodology. Once a phase is completed, it is difficult to revisit or make changes without affecting subsequent phases because any changes usually involve revisiting previous phases and updating documentation, timelines, and resource allocations accordingly. Meanwhile, it is opposite to agile methodology which is more flexible. The project planning is continuously allowing teams to adjust priorities, allocate resources, and refine estimates based on the latest information.

→ Time management

For traditional methodology, it takes a long time because there are fixed timelines established during the planning phases. Plus, each phase must be completed before moving on to the next. Not like agile methodology, it is a lot faster. It is because it has iterative development cycles

Conclusion:

To conclude, we can see that agile methodology gives a big impact to project management nowadays. As the team members are scattered across different time zones and different locations, agile methodology makes the way easier and takes over the traditional methodology.

Question 4

- (a) Calculate the cost-benefit estimation using the Present Value (PV) analysis (for 3 years) to assess the economic feasibility for the SkorBestari project proposal in the worksheet.

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
• Hardware	50000			
• Software	15000			
• Training	30000			
• Consulting	80000			
• Data Conversion	40000			
Total	215000			
Production Costs				
• Supplies		15000	15000	15000
• IS Salaries		75000	82500	90750
• Upgrades		10000	10000	10000
Annual Production Costs		100000	107500	115750
Present Value		74074.07	58984.91	47045.67
Accumulated Costs (Present Value)		289074.07	348058.98	395104.66

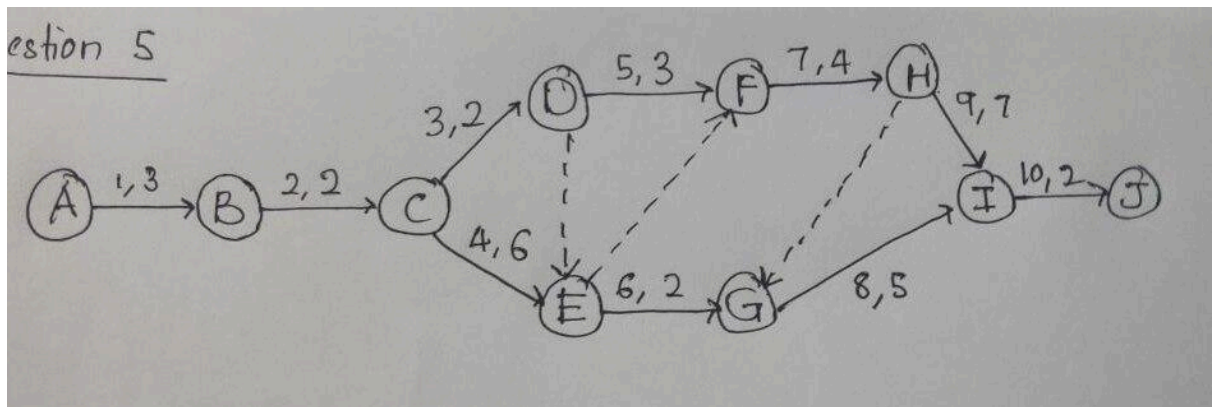
Benefits	Year 0	Year 1	Year 2	Year 3
•Improve Customer Service		150000	180000	216000
•Increase productivity		150000	187500	234375
Annual Benefits		300000	367500	450375
Present Value		222222.22	201646.09	183051.36
Accumulated Benefits (Present Value)		222222.22	423868.31	606919.68
Gain or Loss		-66851.85	75809.33	211815.02
Profitability Index	0.9852			

- (b) What is the profitability index (PI) value for this PV analysis? What is your recommendation based on the PI value and justify it.

The profitability index (PI) value is 0.9852. This is not a good investment because the PI value is below 1.

QUESTION 5

- a. Draw a PERT Diagram for the activities.



- b. Lists all the paths.

1. A - B - C - D - F - H - I - J
2. A - B - C - D - E - G - I - J

- c. Calculate and identify the critical path.

1. A - B - C - D - F - H - I - J
 $= 3+2+2+3+4+7+2$
 $= 23$
2. A - B - C - D - E - G - I - J
 $= 3+2+6+2+5+2$
 $= 20$

Critical path is path 1 (A - B - C - D - F - H - I - J)

- d. Construct a Gantt chart for the project defined above.

